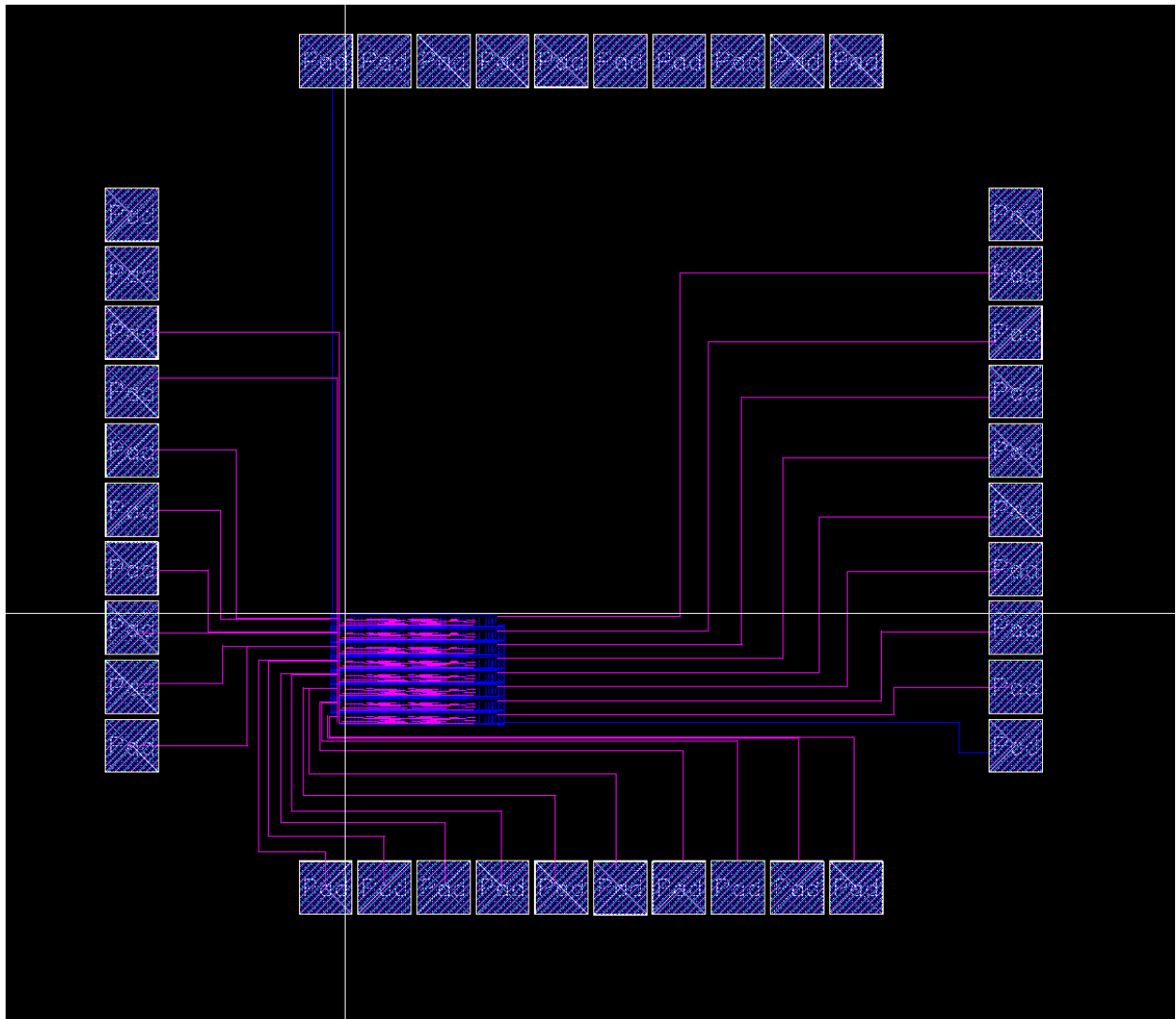


Xor Encryptor/ Decryptor

CPEG 460 -- Introduction to VLSI systems --Project 2- Final Report
Instructor: Prof. Lazarus



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Introduction

This system is an 8 bit encryption/decryption device that uses the exclusive -or method for encryption. It takes as input both an 8 bit data set and an 8 bit encryption key array, both of which the user is given free reign to choose any value for. The entire system consists of the initial encryptor, followed by memory storage to retain both the encrypted data and the key used at the time of encryption. The memory storage then passes the data to the display stage. In the display stage a decryptor is used alongside a multiplexer. Both the encrypted value and the key are passed to the decryptor to reveal the initial data entered, which is then sent to the 2-to-1 multiplexor file. Simultaneously the full hidden value directly from the memory unit that stores it is also passed to the mux file, and the user is presented with the choice to reveal the obscured data or the data previously decrypted.

The entire system is clocked to ensure that each key- data pair is always stored and used together, so that the initial data is never lost while being encrypted.

Specifications

I/O Table:

Type	Input Names							
Xor_key	k0	k1	k2	k3	k4	k5	k6	k7
User_Input	i0	i1	i2	i3	i4	i5	i6	i7
Selector	sel							
clock	clk							

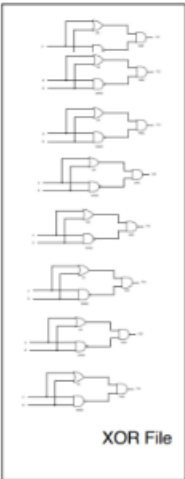
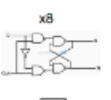
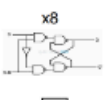
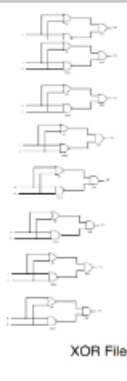
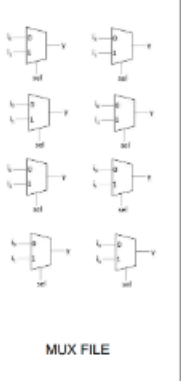
Type	Output Names							
Mux_out	m0	m1	m2	m3	m4	m5	m6	m7

Operation Theory:

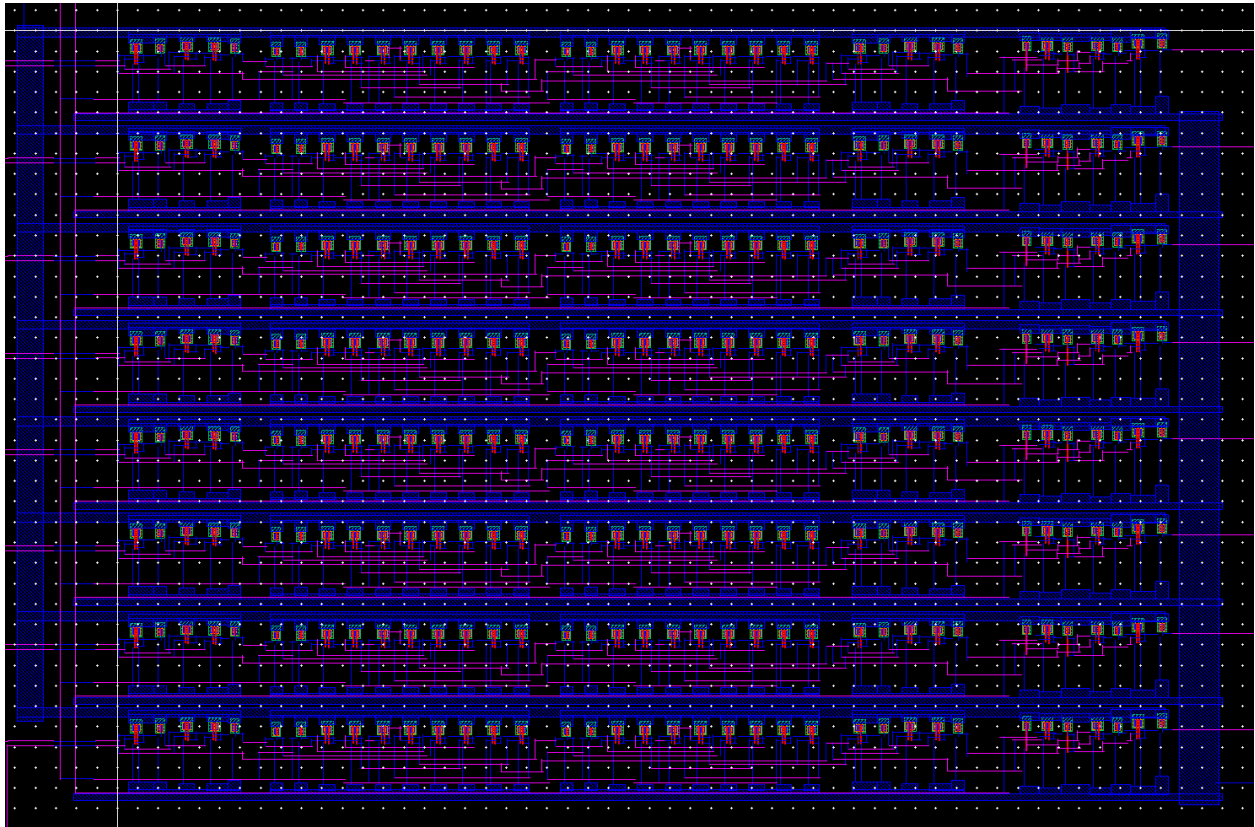
In total there are 18 inputs and 8 outputs. The first 8 input K0 through K7 correspond to the key used to encrypt the data entered by the user. Each K input corresponds to each i input of their same respective number. Inputs i0 through i7 are the data bits that are entered alongside the key. The clk input is used to synchronize the sequential logic of the memory storage component. The Selector input can be set to 0 or 1 at any time to reveal the contents of the mux file, which at anypoint in time holds both the unencrypted and encrypted data which can be selected via 0 and 1 respectively. The 8 outputs m0 through m7 depend entirely on the value of the sel input. When sel = 0 outputs m0-m7 will directly represent inputs i0-i7. When sel = 1 , an xor'd version of i0-i7 will be seen on m0-m7 respectively.

Floor Plan

Initial Floor Plan

Encryption Section	Storage Section	Decryption/ display Section
	  ENCRYPTED DATA FILE XOR KEY FILE	 
8 - parallel xor file Roughly (W x L) 20 x 104	2 - 8 bit register files side by side, utilizing D - flip flops Roughly(W x L) (20 x 104) each So (40 x 104)	1 duplicate xor file followed by an 8 2-to-1 mux file. The mux file is roughly (30 x 104) So together (50 x 104)

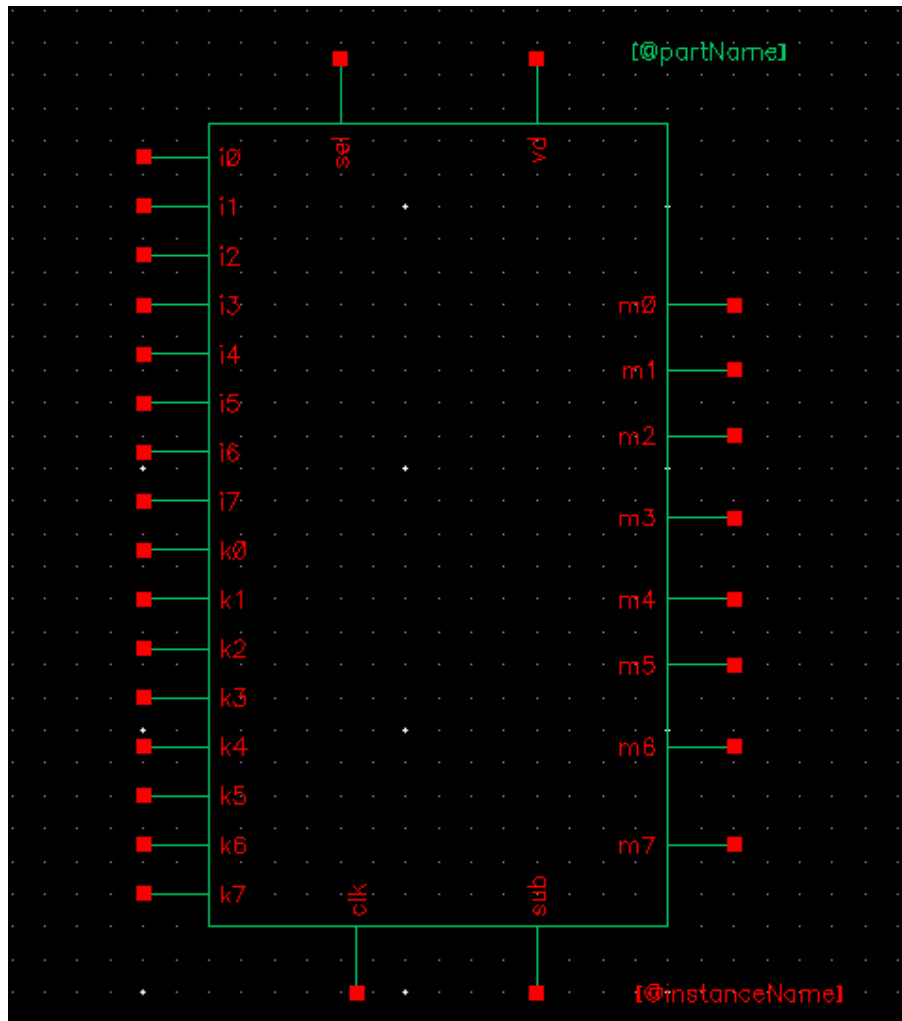
Final Layout



The final layout and the initial floor plan resemble one another almost exactly. The positioning and sectioning of the three main stages remained mostly the same. The dimensions of each portion probably become slightly larger to make room for routing wires and the buses for vd and sub make the layout slightly wider than initially considered but overall the two are comparable.

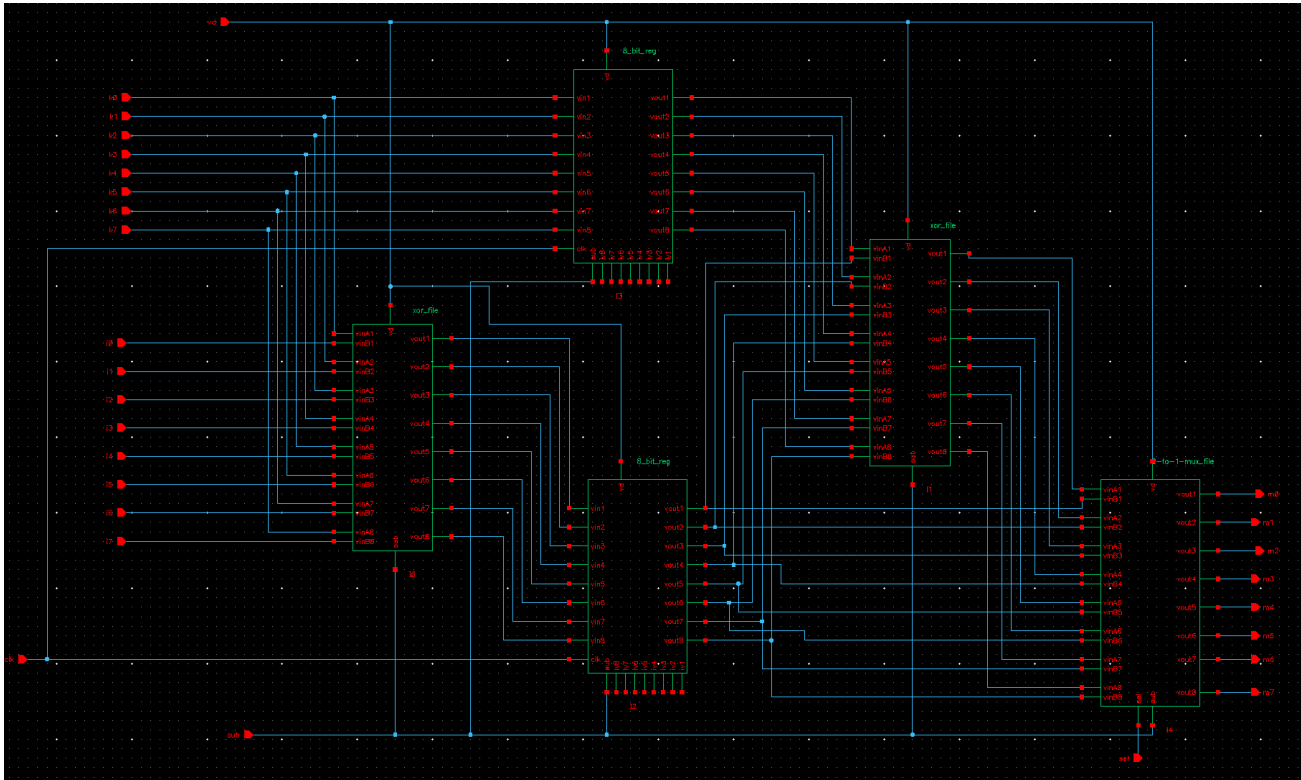
An attempt at a more compact floor plan that reduced the layout width by at least half and reduced length as well. However this floorplan had issues passing verification.

Pinout diagram and symbol for the encryptor:

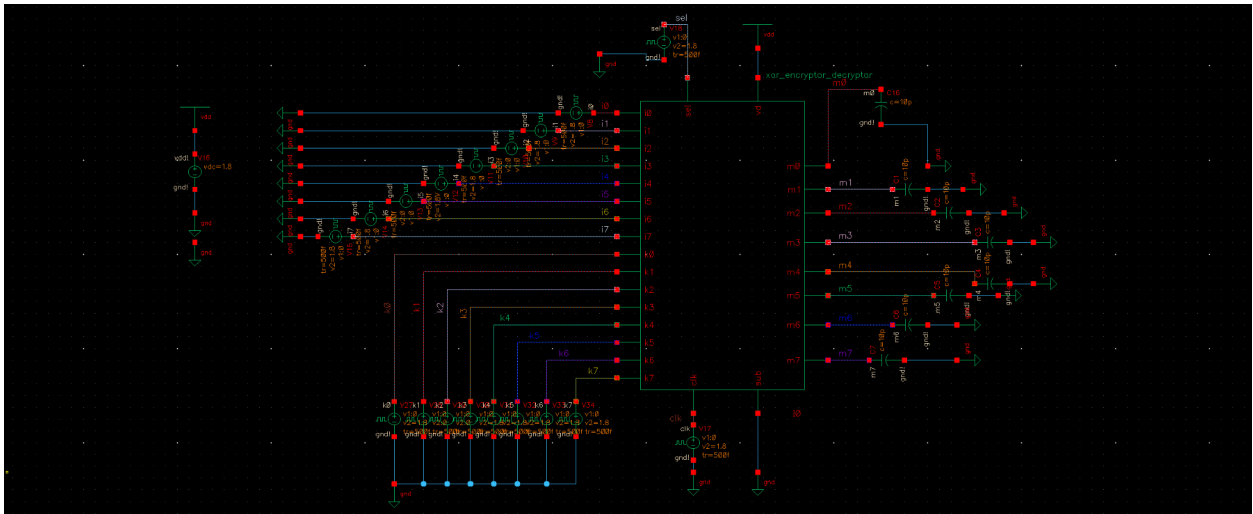


Verification

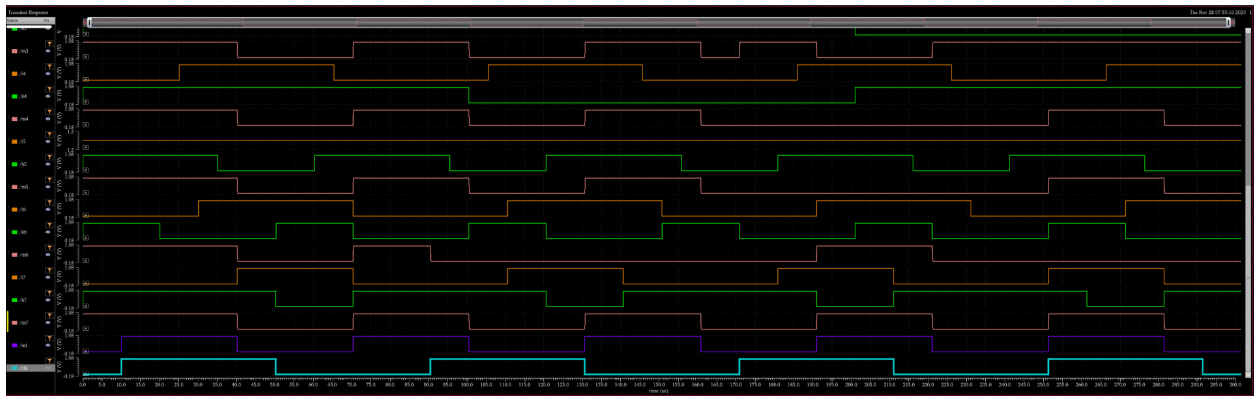
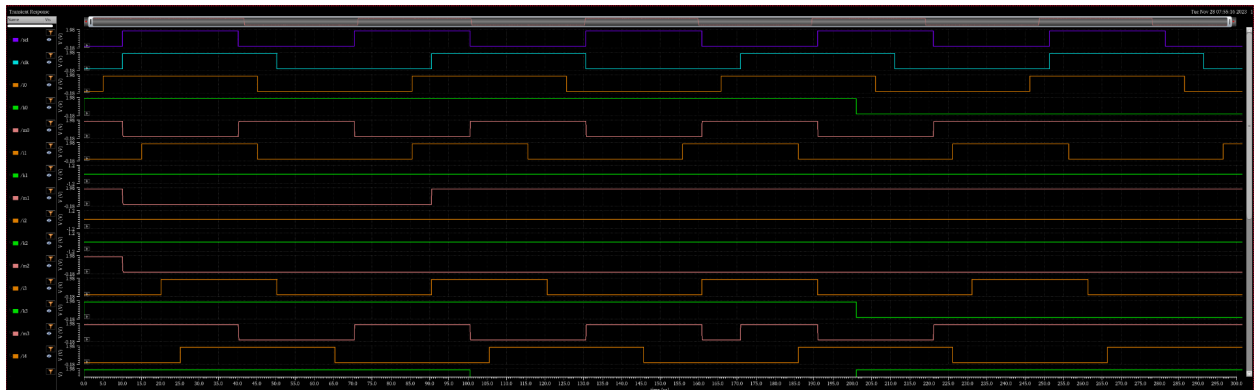
Schematic:



Testbench Schematic:



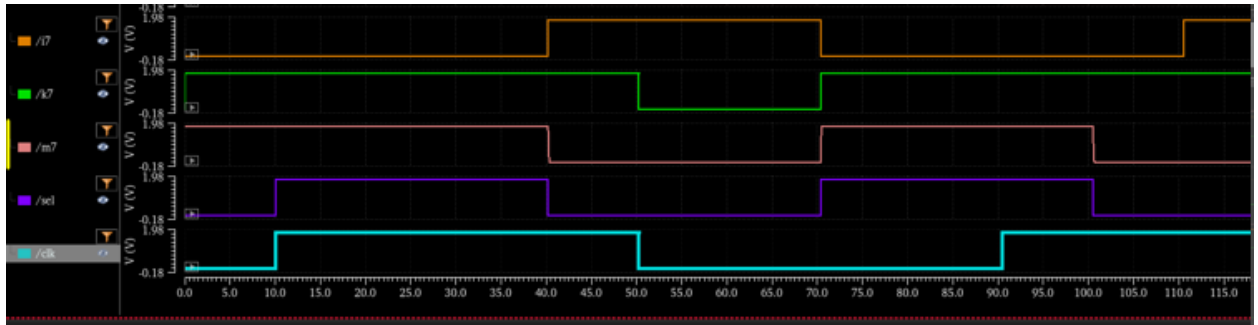
Simulation Result:



The values of i and k are stored in their respective register on the rising clock edge. The outputs m always depict the previous stored values

The table explains how to interpret the data.

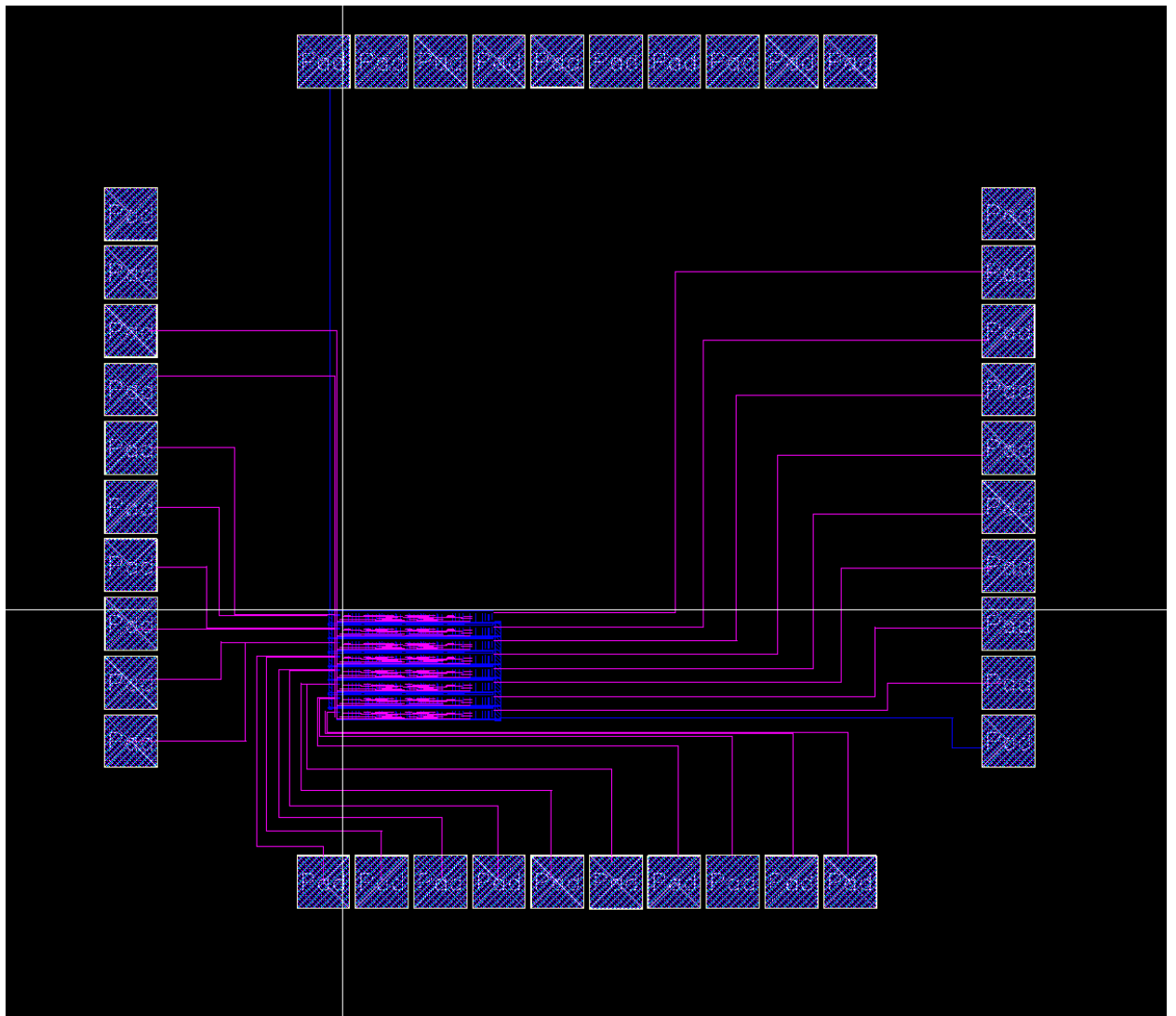
Sel	M outputs
0	Displays i values stored on the previous rising clock edge
1	Displays the xor content of i and k stored on the previous rising clock edge



Sim Results: of k7, i7 , m7	
Time	Signals
t=10	i and k are stored i=0 k=1 Sel = 1 so m retains 1 value the xor of 0 and 1
t=40	sel = 0 The current value of i = 1 Stored value of i = 0 m = 0 and reflects the previous value of i
t=70	sel = 1 m = 1, the previous stored xor value as clock has not risen to accept any new input

The simulation results provided expected values and the circuit is working correctly.

Layout:



Drc and Lvs:

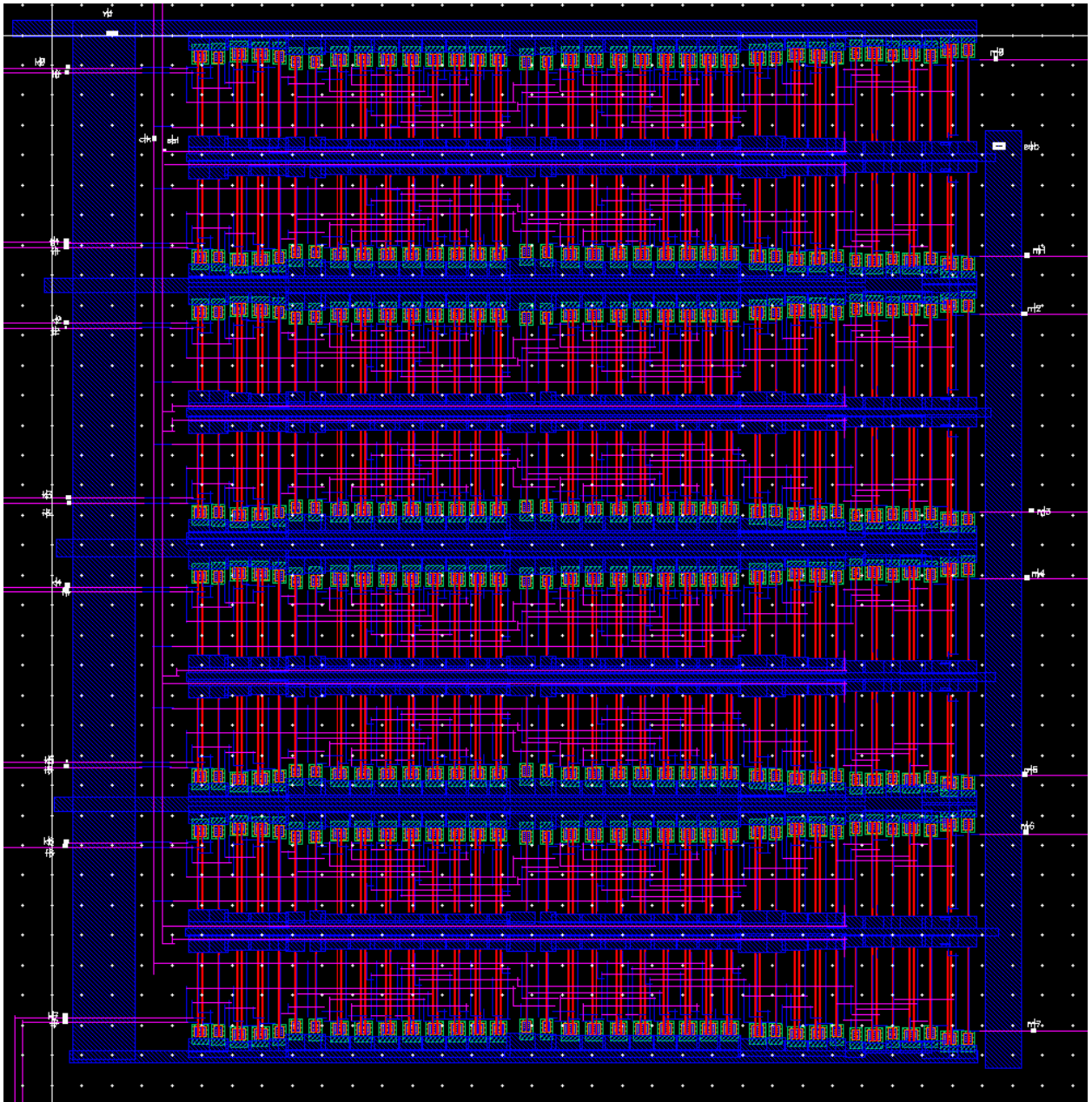
```
DRC started at Wed Dec 6 21:32:13 2023

Validating hierarchy instantiation for:
library: Stanford
cell:    xor_encryptor_decryptor
view:    layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaDRC.rul.
Inclusion limit is set to 1000.
Running layout DRC analysis
Flat mode
Full checking.
DRC started.....Wed Dec 6 21:32:13 2023
completed ...Wed Dec 6 21:32:14 2023
CPU TIME = 00:00:00 TOTAL TIME = 00:00:01
***** Summary of rule violations for cell "xor_encryptor_decryptor layout" *****
Total errors found: 0
```

Total errors found: 0

```
saving rep Stanford/xor_encryptor_decryptor/extracted
Getting layout proper bagGetting layout proper bag
LVS job is now started...
The LVS job has completed. The net-lists match.
```

Attempted Second layout:



Failed Lvs:

```
Run Directory: /home/eceguest/digital_course/LVS
LVS job is now started...
The LVS job has completed. The net-lists failed to match.

Run Directory: /home/eceguest/digital_course/LVS
*WARNING* (LE-104054): Stretch expansion is limited to 5000 wiring objects. Conne
maintained.
```

Troubleshooting output.

8 net-list ambiguities were resolved by random selection.

The net-lists failed to match.

	layout	schematic
	instances	
un-matched	48	48
rewired	16	0
size errors	0	0
pruned	0	0
active	992	992
total	992	992
	nets	
un-matched	16	24
merged	0	0
pruned	0	0
active	508	516
total	508	516
	terminals	
un-matched	0	0
matched but different type	0	0
total	28	28

While the initial layout was designed correctly the second layout would have provided a more compact design and fixed the issue of the nmos in the gates being too far from their well taps. The second layout could be implemented but more time should be spent, designing the individual component gates with these fixes in mind.

Post Fabrication test plan

Input k0-k7 ; i0-i7; and the sel could be wired to simple switches to provide their binary inputs. The outputs m0-m7 could be wired to an LEDs for simplicity but a better plan may be to use an led number display to visually see the distinctive change of the encrypted vs the initial values.

Clk could be connected to a waveform generator to produce a relatively slow periodic signal to act as the clocking mechanism.

Design Time

Floorplan design, including system function, size	3-4 hours
Schematic design and simulation design to test the system	6-8 hours
Block layout and pass DRC & LVS.	3 days
Chip layout design and test.	4 -5 hours